Notes for Prof. Koichi Takahashi

Yusuke Ozaki
Work at RIKEN BDR, Laboratory for Biologically Inspired Computing Updated: November 11, 2026

Thank you for agreeing to write for me. This memo collects facts—periods, what I worked on, and episodes with dates—to help you recall our work. It is not a draft letter; what to evaluate and how is entirely yours. Details and figures: ozatamago.github.io/research; CV: ozatamago.github.io/assets/cv.pdf.

Our working relationship

- Research Part-timer II, Jun 2024 – Aug 2025 and Jun 2026 – present (the gap was my exchange year at the University at Albany, SUNY). Also in the RIKEN AGIS program since Jun 2026.
- Day-to-day collaboration with Dr. Kazunari Kaizu.
- How we interacted: weekly lab-wide progress meetings and a monthly one-on-one meeting; you also reviewed my manuscripts before submission. In the one-on-ones we usually discussed my plan for the next month, my career path, and the larger research direction I will need—and how it fits with the strengths of your lab. We have talked more often since 2026.

OFML and ProtocolUCB: what was given and what I did

Given: the lab's question of whether experimental outcomes can be predicted from protocols written in the lab's Object-Flow Programming Language.
My part: the rest of the project, from conceiving the method to writing the papers:

- **Method.** Proposed OFML, which learns a model for each type of experimental operation and combines them, so that it can predict the outcome of a new protocol built from known operations, using only the final measurements.
- **Theory and simulations.** Gave theoretical guarantees for learning the model and for choosing the next experiment (ProtocolUCB), and confirmed in simulations that sharing operation models across protocols helps.
- **Now:** revising the OFML manuscript and preparing an arXiv version with additional experiments.
- **Roles of others.** Dr. Kaizu: day-to-day discussion, the Object-Flow Programming Language (which he develops), advice on implementation, and revisions of the manuscripts. You: posed the research question, advised on direction, and commented on the manuscripts. Other researchers in the lab also guide me day to day on the *E. coli* experiments and on OFML.

Laboratory automation (Jun 2026 – present)

- **Problem and design.** Set the goal of reducing the GUI-specific know-how each instrument needs; designed a system in which an AI agent turns experimental goals given in chat into procedures the instrument can execute, with a human approving each run.
- **Infrastructure.** Had the agent analyze the instruments' control software, checked the results against the real instruments, and led the development of software that lets the agent operate the Tecan Fluent and its attached devices, the Maholo LabDroid, and a Franka robot arm (code: tecan-fluent, inheco-odtc).
- **Supervised tests.** Basic operations on the Fluent (dispensing, plate transport, thermal cycling) and four short protocols on the Maholo.
- **Now:** automating *E. coli* transformation on the Fluent.
- The agent wrote much of the code under my direction, and a person supervised every run on the instruments.

Episodes you may remember

1. **2024 — where OFML started.** The task was to build machine learning that works on protocols, based on the lab's Object-Flow Programming idea. After trying graph neural networks and Gaussian processes, I saw that treating a protocol as a single identifier or a flat feature vector cannot share knowledge of operations across protocols, and shaped OFML as a composition of per-operation models.
2. **Around Jan–Feb 2025 — a remark at a BioDX meeting.** When computational models that predict biological behavior came up, I pointed out that frameworks writing fixed symbols such as genes and their relations as equations may leave out weak interactions between molecules.
3. **From Jun 2026 — when zero-shot transfer failed.** I had expected operation models learned from final measurements alone to transfer directly to unseen protocols, but in experiments the transfer often failed. I

designed stress tests (unseen operations, shifts in what an operation means, poor input coverage, model misspecification, higher-order interactions) to isolate when it fails, stated the conditions under which transfer can work as the limit of the claim, and turned the project toward a new question: actively adding measurements on the target protocol to correct the prediction. I also handled the reproducibility work myself (random seeds, separating pilot and main runs, isolating test data, logging and auditing failed runs).

4. **Around Aug 2026, when the OFML manuscript was submitted — generalizing the theory.** Instead of stopping at a guarantee tailored to OFML, I reformulated the consistency of active learning as a more general problem (function-valued Bayesian active learning), proved the main results, and placed OFML's active correction as a consequence.
5. **Aug 2026 — first closed loop on the Fluent.** Active learning with OFML on colored water: 11 rounds of experiment, measurement, and model update. I showed a video of it at a lab-wide progress meeting.
6. **Around Aug 2026 — planning the next project.** After the OFML manuscript was submitted, we discussed a next project: tuning the parameters of systems-biology differential-equation models with Bayesian optimization and robotic lab automation. Since then I have been studying E-Cell and differential-equation models (the project itself is still at the planning stage).
7. **Talks on this work:** RIKEN BDR Student Symposium (oral, May 2025); AutoML 2025 (poster, Sep 2025); SPIRAL summer school (poster, Aug 2026); YAML 2026 (oral). I shared the slides with you on Slack.

Outputs

- **[R1]** Object-Flow Machine Learning: Operation-Level Surrogate Modeling for Self-Driving Laboratories. Y. Ozaki, K. Kaizu, K. Takahashi. Under review.
- **[W1]** Bayesian Optimization of Protocols. Y. Ozaki, K. Kaizu, K. Takahashi. NeurIPS 2026 Workshop on AI for Drug Discovery (accepted).

How the three letters fit together

Schools read the three letters side by side, so it helps if each covers what that recommender saw most closely. My suggestion, which you are free to ignore:

- **Prof. Takahashi:** the main research (OFML, ProtocolUCB) and the lab automation—taking a research question to a method, theory, implementation, and experiments on physical instruments.
- **Prof. Inokuchi:** computer-science foundations and growth through undergraduate research (sparse graph neural networks, the undergraduate thesis).
- **Prof. Liu:** mathematical research (problem formulation, proofs, re-examining lower bounds) and research carried out in English.

Please describe only what you observed directly. My work with IBM (SPIN) and at the University of Osaka is in the CV as background; I am not asking anyone to evaluate it.

If you adjust by program (optional)

Your letter is the closest view of my experimental and systems work. Most of the programs are in bioengineering, chemical engineering, or the life sciences (BU CompE, Columbia BME, UCSB BioE, Berkeley CBE, UC Riverside CEE, NYU BME, Northwestern IBiS, Cornell Food Science); for these, the link from experimental goals to models and instrument control. For Stanford CS, the originality of OFML's design and its theory. The facts stay the same either way.